# PRINCE VATS

ENVIRONMENTAL ENGINEER

## CONTACT


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Haridwar, India

LinkedIn Profile


## TECHNICAL SKILLS

- Programming Languages:
  - **Python, C, C++**
- MS office Suite
- Deep Learning & AI
  - **CNN, GNN, LSTM, etc.**
- Computational Software:
  - **MATLAB, R, ArcGIS, etc.**
- Air Quality Monitoring & Analysis

## LANGUAGES

- English (Fluent)
- Hindi (Fluent)
- Japanese (Basic)

## PROFILE

Studying Aerosol Science with a focus on applications in environmental engineering, including air-pollution monitoring, spatio-temporal modelling, and low-cost sensor networks. **Achieved 29th rank nationwide (99.56 percentile)** in *India's Graduate Aptitude Test in Engineering (GATE)*—a competitive national exam for engineering graduates. Skilled in developing advanced machine learning models for $PM_{2.5}$ prediction and integrating air quality with meteorological data to generate impactful, research-driven insights.

## RESEARCH EXPERIENCE

### Journal Publication — Published in March 2025

Potential Health Risk Estimation Through Deterministic and Stochastic Approaches for the Air Pollutants Found Inside Bottling Beverage Industries of Himachal Pradesh in India

*(Published in Discover Environment, Springer – https://doi.org/10.1007/s44274-025-00221-x)*

- Assessed indoor air pollution in 12 bottling and beverage industries during the monsoon season.
- Monitored **TVOCs, $PM_{10}$, $PM_{2.5}$, $PM_1$, black carbon (BC), and noise levels.**
- Performed deterministic and probabilistic health risk assessments, identifying non-carcinogenic hazard ratios (HRs) between 0.52–0.80.
- Determined highest risk in males aged 50–60 and females aged 40–50.
- Estimated carcinogenic risk of BC exposure between 5.8E-04 and 6.8E-04.
- Recommended ventilation improvements and targeted mitigation strategies.

### Ongoing Project — ongoing

Environmental Quality Monitoring System Using Low-Cost Sensors for urban Air Quality Monitoring

- Developing monitors using multiple low-cost sensors for monitoring **$PM_1$, $PM_{2.5}$, $PM_{10}$, $CO_2$, temperature, pressure, humidity, noise levels, CO, wind speed, and TVOCs.**
- Deploying dense network across urban area with real-time data transmission to custom made central server.
- Implementing CNN-Bidirectional LSTM model for anomaly detection, spatio-temporal analysis, and forecasting for 24h, 7d, and 1-month horizons.
- Supporting policy-making through real-time urban air quality insights.

### Manuscripts Under Review

Evaluating CNN-LSTM and GNN-GRU Hybrid Models for Enhanced Air Quality Prediction and Dynamic Forecasting

- Compared deep learning models for 24-hour $PM_{2.5}$ prediction: CNN-LSTM vs. GNN-GRU.
- Used hourly $PM_{2.5}$ data from 35 stations in Delhi.
- Achieved RMSE of 26.67 $\mu g/m^3$ with optimized CNN-LSTM.
- Demonstrated improved spatial accuracy using GNN-GRU with reduced computational load.
- Applied feature engineering and PCA to enhance interpretability and performance.

Evaluating Multiple ML Models for AQI Prediction in Gangetic Plain Using Air Quality and Meteorological Features *(under review in Env. monitoring and Assessment)*

- Evaluated 15 ML models for PM2.5-based AQI prediction in 25 Gangetic Plain cities using air quality and meteorological data (2018–2023).
- Extra Trees was the top performer, then Gradient Boosting, LightGBM, and XGBoost.
- Linear models and SVR showed higher errors, especially in high-variability cities like Lucknow and Muzaffarpur, due to the inability to capture nonlinear AQI-meteorology links.
- Ensemble models consistently gave lower errors, stronger generalization, and stable residuals. Findings highlight ensemble ML's superiority for accurate, city-specific AQI forecasting and environmental decision-making.

## CO-CURRICULAR ACTIVITIES

- Taught underprivileged students through an NGO initiative
- Acted in college dramas.
- Acted in college dramas.
- Managed cultural fest events.
- Vice President, Green Club.
- Math & Science Olympiads.
- Volunteered in tree plantations.
- Attended environmental workshops.
- Organized technical seminars.
- Joined hackathons.
- Assisted in sensor demos.
- Community clean-up drives.

## EDUCATION

**Master of Technology** 2022 - 2024

Indian Institute of Technology Ropar

**About thesis:**

- Objective: Investigated indoor air pollution levels and explored the role of indoor plants in mitigating fine particulate matter (PM).
- Air Quality Monitoring:
  - Monitored $PM_{10}$, $PM_{2.5}$, $PM_1$, and Black Carbon (BC) across classrooms, hostels, and office spaces using real-time instruments.
  - Found $PM_{2.5}$ (59.6 μg/m$^3$) and $PM_{10}$ (71.12 μg/m$^3$) concentrations exceeding CPCB (Central Pollution Control Board) permissible limits, highlighting significant indoor exposure risks.
- Plant-Based Intervention:
  - Tested the Ajwain plant (Trachyspermum ammi) and the Money plant (Epipremnum aureum) for their particulate matter reduction capacity.
  - Results showed:
    - Ajwain plant: $PM_{2.5}$ reduction by 39.3%.
    - Money plant: $PM_{10}$ reduction by 38.7%.
  - Proposed a practical mitigation strategy: placement of 20 Ajwain plants or 30 Money plants per room to meet CPCB indoor air quality standards.
- Health Assessment:
  - Conducted spirometry tests on 303 students to evaluate the respiratory health impact of poor indoor air quality.
  - Key findings:
    - 35.97% showed restrictive lung disease symptoms.
    - 34.65% showed obstructive patterns.
    - 74.59% showed asthma prevalence, demonstrating strong correlation between indoor PM exposure and respiratory health issues.
- Significance:
  - First systematic study at IIT Ropar linking indoor air quality, plant-based interventions, and respiratory health outcomes.
  - Provides a low-cost, eco-friendly strategy for improving indoor air quality in schools, offices, and households.

**Bachelor of Technology (Civil Engineering)** 2018 - 2022

GBPUAT, Pantnagar

## RESEARCH EVENTS ATTENDED

1. Dec 2025: Symposium on Advances in Tools and Technologies to Quantify Health Impacts of Air Pollution, at IIT Gandhinagar.
2. Dec 2025: Asian Aerosol Conference (AAC 2025), organized by IASTA, Mumbai – One presentation and one poster presented.
3. Aug 2025: India Clean Air Summit 2025, organized by CSTEP, Bengaluru – Two poster presentations submitted.
4. Mar 2025: Smart Sensors and Data Analytics for Air Quality Management Workshop, IIT Bombay – Participant.
5. Dec 2023: International Conference on Advances in Air Quality Management (IICAQM 2023), IISc, Bengaluru – One paper presentation (https://doi.org/10.1007/978-981-96-2359-4_18).
6. Aug 2023: AI4Water: Advanced Data-Driven and Artificial Intelligence Tools in Water Resources Workshop, IIT Roorkee – Participant.
7. Aug 2023: 7th National Water Seminar, National Institute of Hydrology, Roorkee – Two paper presentations; Best Paper Award received.

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## EDUCATION

**PhD Candidate, Env. Engineering** Joined in 2024
Indian Institute of Technology Gandhinagar

**Doctoral Research (Ongoing):**

• AI-driven spatio-temporal analysis and prediction of urban air quality using dense IoT-based environmental monitoring networks
• Development and deployment of low-cost IoT Environmental Quality Monitoring Systems (EQMS) for high-frequency urban data collection
• Machine learning–based sensor calibration, data fusion, and exposure modeling
• Deep learning models (GNN, LSTM/GRU, CNN–BiLSTM) for short-term and day-ahead PM2.5 forecasting

**Master of Technology** 2022 - 2024
Indian Institute of Technology Ropar

**About thesis:**

This study investigated indoor air pollution at IIT Ropar and evaluated indoor plants as a mitigation strategy, monitoring $PM_{10}$, $PM_{2.5}$, $PM_1$, and black carbon in classrooms, hostels, and offices using real-time instruments; results showed $PM_{2.5}$ (59.6 μg/m$^3$) and $PM_{10}$ (71.12 μg/m$^3$) levels exceeding CPCB limits, indicating significant exposure risks. The Ajwain plant (Trachyspermum ammi) and Money plant (Epipremnum aureum) were tested for particulate reduction, with Ajwain reducing $PM_{2.5}$ by 39.3% and Money plant reducing $PM_{10}$ by 38.7%, leading to a practical recommendation of 20 Ajwain or 30 Money plants per room to meet CPCB standards. A health assessment using spirometry on 303 students revealed 35.97% with restrictive lung disease, 34.65% with obstructive patterns, and 74.59% asthma prevalence, demonstrating a strong link between indoor particulate exposure and respiratory health issues. This first-of-its-kind systematic study at IIT Ropar integrates air quality monitoring, plant-based intervention, and health outcomes, offering a low-cost, eco-friendly solution to improve indoor air quality in educational, office, and residential settings.

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GBPUAT, Pantnagar

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